



FEU INSTITUTE OF TECHNOLOGY

COMPUTER SCIENCE DEPARTMENT

CS0019

Modeling and Simulation

TECHNICAL ASSESSMENT

1

Introduction to MATLAB

Student Name:	Jecyn Vallirie M. Turbanos
Section:	TN36
Professor:	Jericho Menguito



DevNet Resource Validation

Use AI to recommend a DevNet resource, verify it against official Cisco documentation, and record the plan in Git.

MODULE 2	DURATION 90 minutes	MODE Individual or Pair	CHECKS 9
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Core rule

An AI recommendation is provisional. Accept it only after an official Cisco source supports the resource type, access model, and intended use case.

1. Overview

You are preparing a development-environment plan for four fictional network-automation tasks. For each task, you will ask an approved AI tool to recommend one DevNet resource category, independently verify the recommendation using official Cisco documentation, complete a structured JSON plan, run an offline validator, and preserve the evidence in a local Git repository.

2. Learning outcomes

- Explain the roles of DevNet Learning Labs, DevNet Sandbox, and Cisco Code Exchange.
- Differentiate Always-On and Reservation Sandboxes based on access, isolation, setup, and administrative privileges.
- Select and justify an appropriate DevNet resource for a stated development or automation need.
- Validate AI-generated claims using official technical documentation.
- Use Git to preserve a reproducible resource-selection and validation record.

3. Required tools and files

Item	Requirement
Browser	Access only official Cisco DevNet pages for evidence
AI tool	Instructor-approved generative AI assistant



Item	Requirement
Python	Python 3.10 or later; no third-party packages
Version control	Git installed locally; GitHub publishing only when instructed
Starter files	devnet_use_cases.json, student_plan_template.json, validate_plan.py, README_template.md

4. Safety, ethics, and account protection

Read-only exploration by default

Do not create a reservation, connect a VPN, run API requests, or change a sandbox unless the instructor explicitly authorizes it. Never paste usernames, passwords, API keys, tokens, reservation emails, private repository content, or personal data into an AI tool.

- Use only fictional use cases supplied with the activity.
- Official evidence must come from developer.cisco.com.
- Record whether each AI recommendation was verified, partially verified, or rejected.
- A correct selection without evidence and AI disclosure is incomplete.

5. Resource decision guide

Resource	Use when	Important constraint
Learning Lab	Guided, step-by-step learning is the main need	Learning content; not a substitute for choosing an execution environment
Always-On Sandbox	Immediate shared access and non-admin API exploration are sufficient	Shared environment with restricted administrative access
Reservation Sandbox	Private access, administrative control, or configuration testing is required	Reservation, setup time, and often VPN access are required
Cisco Code Exchange	Existing code repositories or automation examples are needed	Code must still be reviewed, tested, and matched to the use case

6. Official evidence sources

- DevNet Sandbox documentation: <https://developer.cisco.com/docs/sandbox/>
- DevNet Learning Labs: <https://developer.cisco.com/learning/>



- Cisco Code Exchange: <https://developer.cisco.com/codeexchange/>

Terminology update

Cisco states that Automation Exchange has been consolidated with Code Exchange. Use Cisco Code Exchange in the revised syllabus and student outputs.

7. Procedure

1. **Prepare the workspace:** Create a folder named `it0123-devnet-resource-plan` and place all starter files inside it.
2. **Initialize Git:** Run `git init`, add `README_template.md` as `README.md`, and create an initial commit with a meaningful message.
3. **Inspect the scenarios:** Open `devnet_use_cases.json`. Identify the decisive requirement in each use case before consulting AI.
4. **Prompt AI:** Ask the AI to recommend one of four resource types: `learning-lab`, `always-on-sandbox`, `reservation-sandbox`, or `code-exchange`. Require a reason and a claim that can be verified.
5. **Verify independently:** Open the relevant official Cisco page. Check the AI claim against the resource description, access model, and intended use.
6. **Complete the plan:** Copy `student_plan_template.json` as `student_plan.json`. Fill every field using the exact resource labels required by the validator.
7. **Run validation:** Execute `python validate_plan.py`. Correct the plan until the validator reports 9/9 checks passed.
8. **Complete the README:** Summarize the decisions, official evidence, AI evaluation, validator result, and AI-use disclosure.
9. **Commit the evidence:** Add the completed JSON, README, and a text copy or screenshot of the validator result. Commit with a message that describes the validation milestone.
10. **Prepare for oral checking:** Be ready to defend one resource decision and explain why an alternative resource was less suitable.



8. Suggested AI prompt

I am selecting a Cisco DevNet resource for a fictional classroom use case.
Use case: [paste one scenario]
Choose exactly one: learning-lab, always-on-sandbox,
reservation-sandbox, or code-exchange.
Explain which requirement drove your choice. State any access, isolation,
setup, or privilege claim that I should verify in official Cisco documentation.
Do not invent a sandbox product name, URL, account, or credential.

9. Required labels

learning-lab
always-on-sandbox
reservation-sandbox
code-exchange

Verification status: verified | partially-verified | rejected

10. Evidence and deliverables

- student_plan.json with all four validated resource decisions

```
{ } student_plan_template.json  
{ } student_plan.json
```

- README.md with official sources, Git evidence, and AI-use disclosure



IT0123 Module 2 DevNet Resource Validation

Resource Decisions

Use case	Final resource	Verification	Main reason
UC-01	learning-lab	verified	The scenario needs guided, step-by-step learning before choosing an execution environment.
UC-02	always-on-sandbox	verified	The scenario needs immediate shared API exploration without admin access, VPN setup, or configuration changes.
UC-03	reservation-sandbox	verified	The scenario needs private isolated configuration testing with administrative control.
UC-04	code-exchange	verified	The scenario needs existing sample code or a repository to review and adapt.

Official Evidence

- Cisco DevNet lists Labs as automation and programmability labs featuring Cisco platforms:
<https://developer.cisco.com/learning/>
- Cisco DevNet Sandbox documentation distinguishes always-on and reservation-based sandbox access models:
<https://developer.cisco.com/docs/sandbox/>

- validator output showing 9/9 checks passed

```
PS C:\Users\Cynthia Molato\Downloads\it0123-devnet-resource-plan> python vali
date_plan.py
PASS: 1/9 plan contains four decisions
PASS: 2/9 all required use cases are represented
PASS: 3/9 resource labels are valid
PASS: 4/9 resource choices match scenario requirements
PASS: 5/9 verification statuses are valid
PASS: 6/9 official evidence uses developer.cisco.com
PASS: 7/9 AI recommendation summaries are recorded
PASS: 8/9 official evidence summaries are recorded
PASS: 9/9 final decision reasons are recorded
```

- local Git repository with at least two meaningful commits

```
Result: 9/9 checks passed
PS C:\Users\Cynthia Molato\Downloads\it0123-devnet-resource-plan> git log --
neline
3e1f957 (HEAD -> main, origin/main, origin/HEAD) Just updated some of here
b549165 Complete DevNet resource validation plan
e01f96a Add DevNet validation activity scaffold
PS C:\Users\Cynthia Molato\Downloads\it0123-devnet-resource-plan>
```

- AI prompt record or transcript excerpt for each scenario



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```
PS C:\Users\Cynthia Molato\Downloads\it0123-devnet-resource-plan> type ai_prompt_record.md
# AI Prompt Record

## UC-01

Prompt: Choose exactly one Cisco DevNet resource type for a beginner team that needs guided
YANG, NETCONF, and RESTCONF practice before choosing an execution environment.

AI recommendation: `learning-lab`

Verification: Accepted after checking the official DevNet Learning Labs source.

## UC-02

Prompt: Choose exactly one Cisco DevNet resource type for immediate read-only API exploratio
n with no admin privileges, topology changes, VPN setup, or private isolation.

AI recommendation: `always-on-sandbox`

Verification: Accepted after checking the official DevNet Sandbox source.

## UC-03

Prompt: Choose exactly one Cisco DevNet resource type for testing configuration automation i
n a private lab with administrative access, where setup time and VPN access are acceptable.

AI recommendation: `reservation-sandbox`

Verification: Accepted after checking the official DevNet Sandbox source.

## UC-04

Prompt: Choose exactly one Cisco DevNet resource type when the main need is an existing auto
mation repository or reusable code example.

AI recommendation: `code-exchange`

Verification: Accepted after checking the official Cisco Code Exchange source.
PS C:\Users\Cynthia Molato\Downloads\it0123-devnet-resource-plan> █
```

- Optional instructor-approved GitHub repository link

<https://github.com/Vallywi/it0123-devnet-resource-plan>

GitHub Repository: <https://github.com/Vallywi/it0123-devnet-resource-plan.git>



11. Reflection questions

1. Which scenario was easiest to classify, and which requirement determined your choice?

UC-04 was the easiest to classify because it clearly needed an existing automation example or reusable code repository. That requirement directly matched Cisco Code Exchange, since Code Exchange is used for finding sample solutions and implementations.

2. Describe one AI claim that needed correction, qualification, or rejection.

One AI claim that needed qualification was that an Always-On Sandbox is best for API tasks. I verified that it is useful for quick shared access, but it is not suitable for every case, especially when the task requires private isolation, administrative control, VPN access, or configuration changes.

3. Why is an Always-On Sandbox not automatically the best option for every API task?

An Always-On Sandbox is not automatically the best option because it is shared and usually limited. It works well for quick API exploration, but API tasks that need private access, administrative privileges, or configuration testing are better suited for a Reservation Sandbox.

4. How did Git improve the traceability of your resource-selection process?

Git improved traceability by recording the activity progress through commits. It showed when the activity scaffold was created and when the completed validation plan was added, making the resource-selection process easier to review and verify.

12. Assessment rubric

Criterion	Points	Evidence
Resource classification	25	Four selections match the scenario requirements
Official evidence validation	25	Accurate Cisco sources and defensible verification
AI evaluation	20	Recommendations summarized and critically accepted, modified, or rejected
Plan and Git workflow	20	Complete JSON/README, passing validator, meaningful commits
Ethics and professionalism	10	No credentials exposed; clear and complete submission
TOTAL	100	

Submission filename: Surname_FirstName_IT0123_M2_DevNetPlan.zip